

018/016 Assura LVS/LPE Deck Usage

PDKD/TSMC

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Switch and Variable Setting

Assura Switches(I)

- **?set (“ZERO_NRDS”)**
 - Turn on this switch to force NRS=NRD=0.
- **?set (“extract_dnwdio”)**
 - Turn on this switch to extract PW/DNW and DNW/PSUB diodes.
- **?set (“NW_RING”)**
 - Turn on this switch to enable NW ring to separate the node from BULK.
- **?set (“spice_extraction”)**
 - Turn on this switch to omit saveProperty about model name for spice extraction.
- **?set (“Skip_Soft_Connect_Checks”)**
 - Turn on this switch to skip soft-connect check.
- **?set (“CAP_1P5”) / ?set (“CAP_2P0”)**
 - Select which capacitance will be extracted. Default is 1.0fF MIM capacitor.
- **?set (“CDL_input”)**
 - Switch for running GDS mode.

Assura Variables

- **scale**

- For 0.18 process, scale = 1.0 ; for 0.16 process, scale = 0.9. Scale factor can be found from spice model card.
- Scale variable relates to the scale factor in shrink process. Please do not change the default value.

Assura LVS/LPE Flow

Assura LVS Flow

● Run LVS :

- Include "source.added" file in your source netlist for subcircuits.
 - ◆ .include source.added
- Prepare a LVS.rsf file for Assura LVS run in batch mode.
 - ◆ avParameters(
 - ?inputLayout ("GDS2" "top_cell.gds") ; specify full path gds name
 - ?cellName "top_cell" ; specify top cell name
 - ?rulesFile "./1p6m/extract.rul" ; specify assura deck
 - ?runName "top_cell" ; specify run name
 - ?workingDirectory "./rundir" ; specify run directory
 -) ; end of avParameters
 - load("./1p6m/compare.rul") ; load compare.rul
 - avLVS()
 - Run Assura
 - ◆ % **assura LVS.rsf**
 - ◆ Files top_cell.cls, top_cell.err and top_cell.csm are LVS result and path check report.

Assura LPE Flow

● Run LPE:

- Prepare a rcx.rsx file For Assura LPE Flow in batch mode.

```

◆ avParameters(
    ?workingDirectory "./rundir"                ; specify run directory
    ?runName "<run_name>"                        ; specify run name
    ?inputLayout ( "GDS2" "<gds_name>" )        ; specify gds name
    ?cellName "<cell_name>"                     ; specify top cell name
    ?viewName "layout"
)
rcxParameters(
    ?rcxSetupDir "."
    ?outputFormat "spice"
    ?output "<output_name>"                    ; specify output name
    ?runName "<run_name>"                      ; specify run name
    .
    .                                           ; customize RC extraction setting
)
avRCX()

```

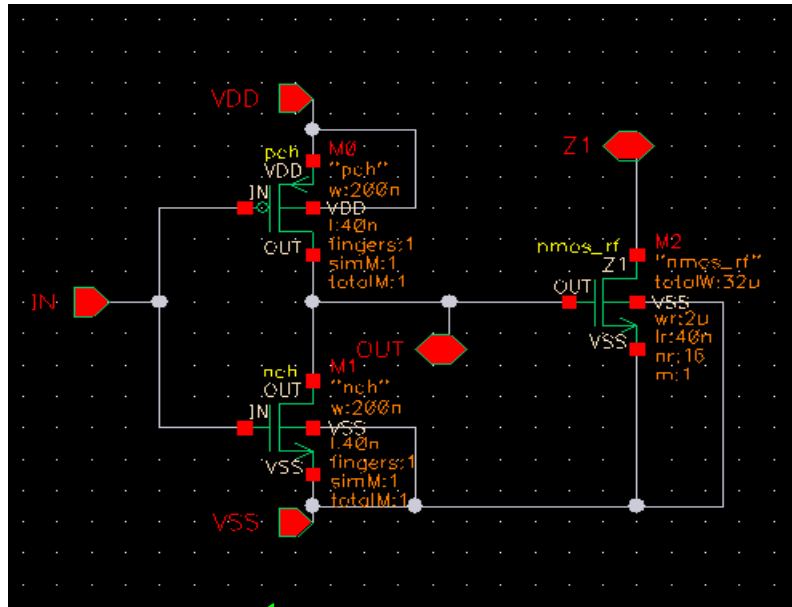
- Run LPE

- ◆ Download (and un-tar) the LPE tech files.
- ◆ Create lvfile under LPE techdir.
- ◆ For Assura LVS/LPE flow, run Assura LVS and copy the runname.xcn file to the LPE techdir and call it lvfile.
- ◆ **% capgen.cmd**
- ◆ **% assura rcx.rsx**

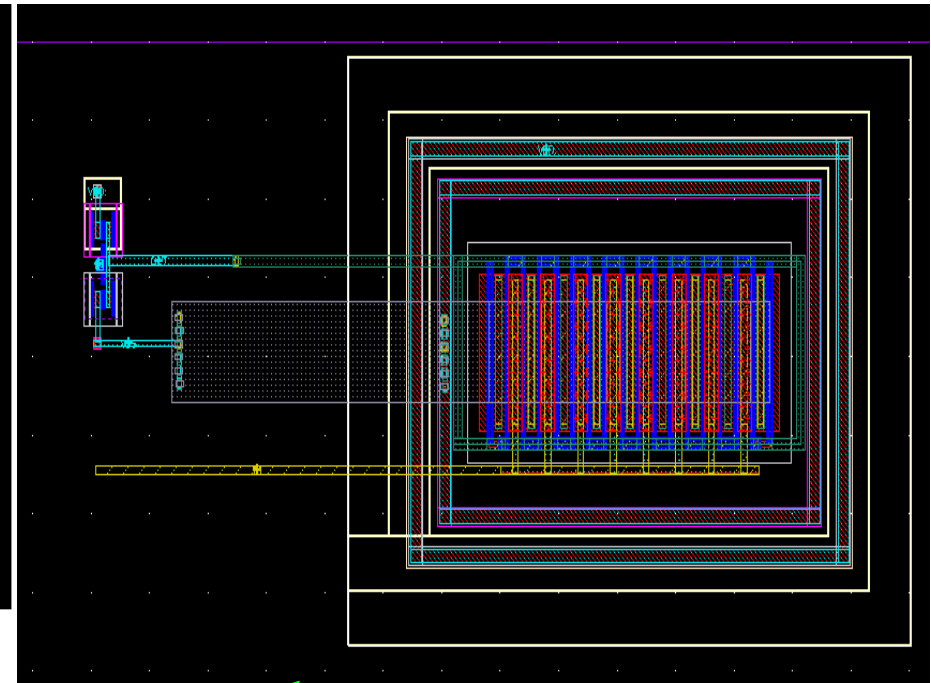
Assura LVS/LPE GUI Flow

Data Preparation

- The library includes schematic view and layout view.
- There are two logic mos devices and one rf device.



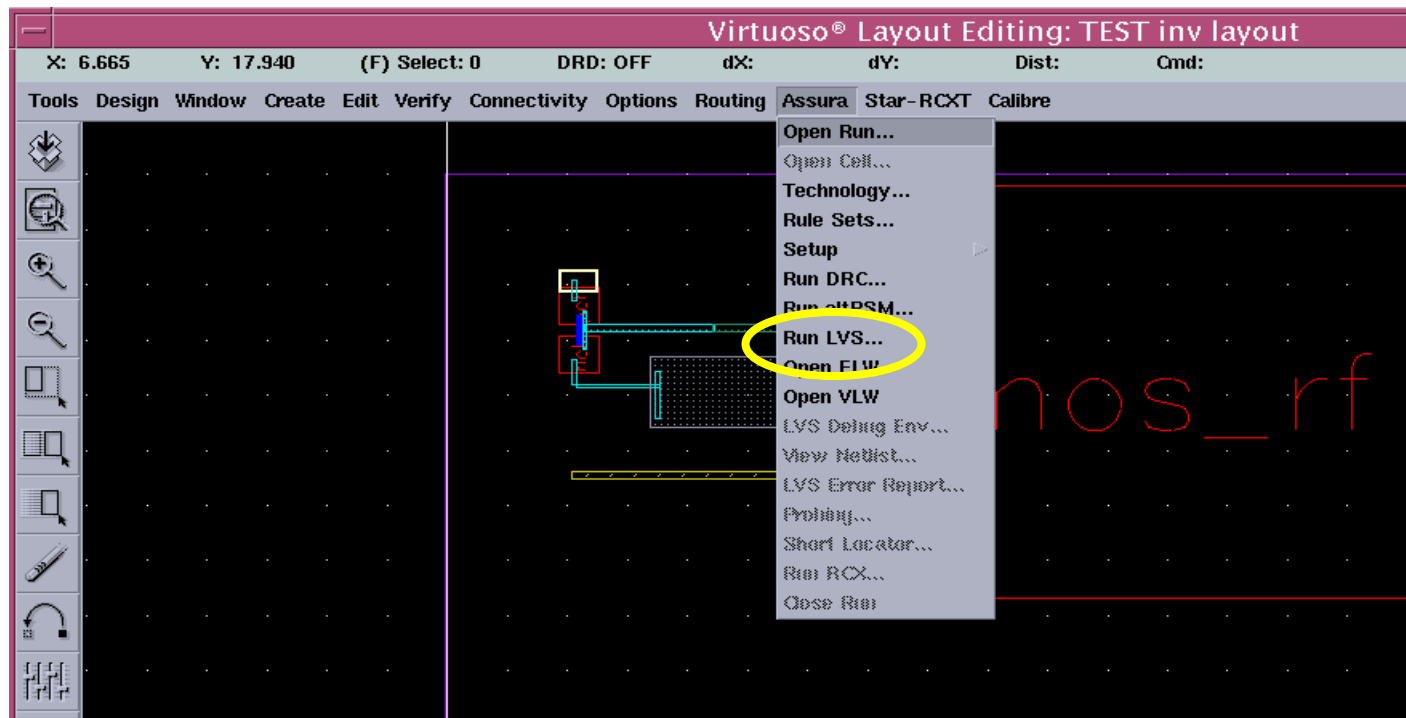
Schematic
view



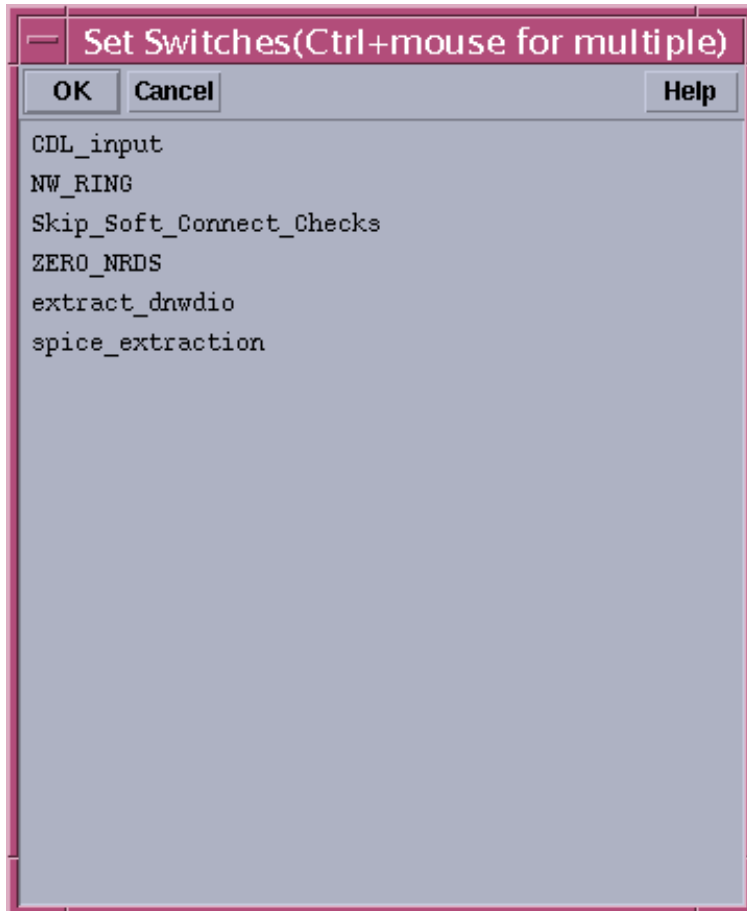
Layout
view

Assura LVS GUI Flow(I)

- Trigger the “Run LVS” in the Assura pull down manual.



Assura LVS GUI Flow(II)



- There is a pop-up manual when the “set Switches” is triggered.
- Switch description
 - **NW_RING**: The NWELL ring could isolate the psub.
 - **Skip_Soft_Connect_Checks**: It will not do the soft-connect check.
 - **ZERO_NRDS**: The NRS/NRD will set to zero which it will take care by the RC extraction.
 - **extract_dnw dio**: Extract PW/DNW, DNW/PSUB diode.
 - **spice_extraction**: When do the spice extraction, this option has to turn on.
 - **CDL_input**: Switch for running GDS mode.

Assura LVS GUI Flow(III)

Run Assura LVS

OK Cancel Apply Defaults Load State Save State View RSF Help

Schematic Design Source **DFII** Use Existing Netlist ☐ Netlisting Options...

Library **TEST** Cell **inv** View **schematic** Browse...

Layout Design Source **DFII** Use Existing Extracted Netlist ☐

Library **TEST** Cell **inv** View **layout** Browse...

Run Name **l** Run Directory **./Intel_run** ...

Run Location **local**

View Rules Files ☒ Technology **assura_tech** Rule Set **default**

☐ Extract Rules **a/PDK_NEW/N45RF/Assura/lvs_rcx/extract.rul** View... Reload

☐ Compare Rules **c/kmliua/PDK_NEW/N45RF/Assura/lvs_rcx/compare.rul** View...

Switch Names **l** Set Switches

☐ Binding File(s) **156b/kmliua/PDK_NEW/N45RF/Assura/lvs_rcx/bind.rul** View...

☐ RSF Include **l** View...

Variable	Value	Default	Description
None			

View avParameters ☒ Modify avParameters... 4 avParameters are set.

```
?joinPins top
?keepOriginal t
?preserveCells ( file("./automount/tsmc156/root/home/tsmc156b/kmliua/PDK_NEW/N45RF/A
```

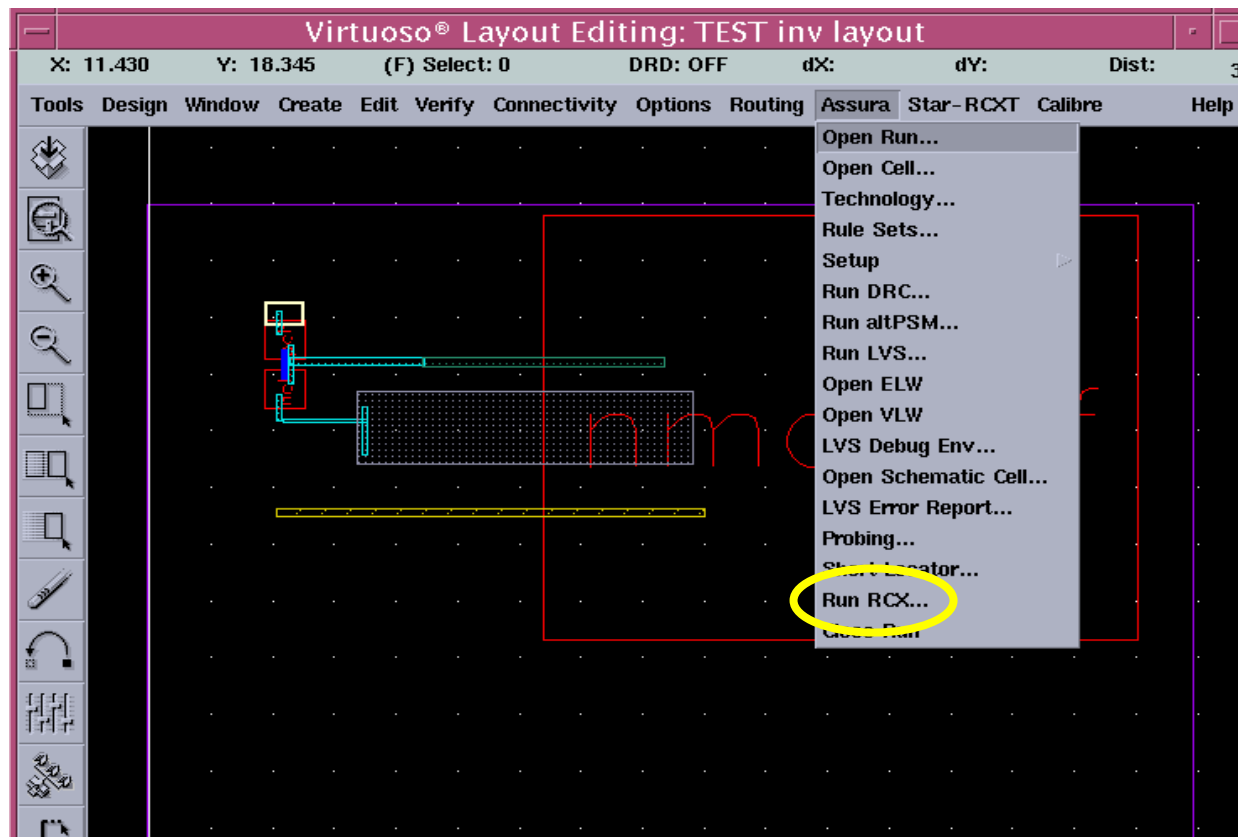
View avCompareRules ☐ Modify avCompareRules... No avCompare rules are set.

View Additional Functions ☐ No additional functions are set.

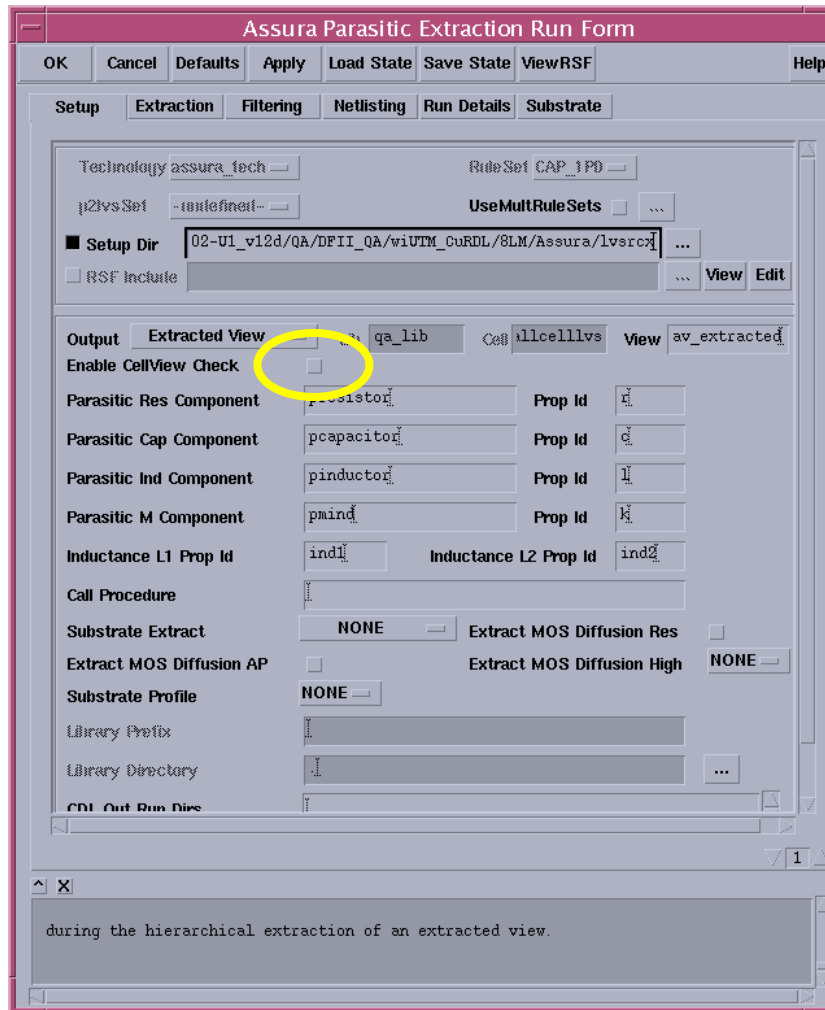
- Turn on “Modify avParameters”.
- The option “?joinPins” means to turn on virtual connect.
- Please check “run_name.cls”, “run_name.err”, and “run_name.csm” files after the LVS check.

Assura LPE GUI Flow(I)

- Trigger the “Run RCX” in the Assura pull down manual.

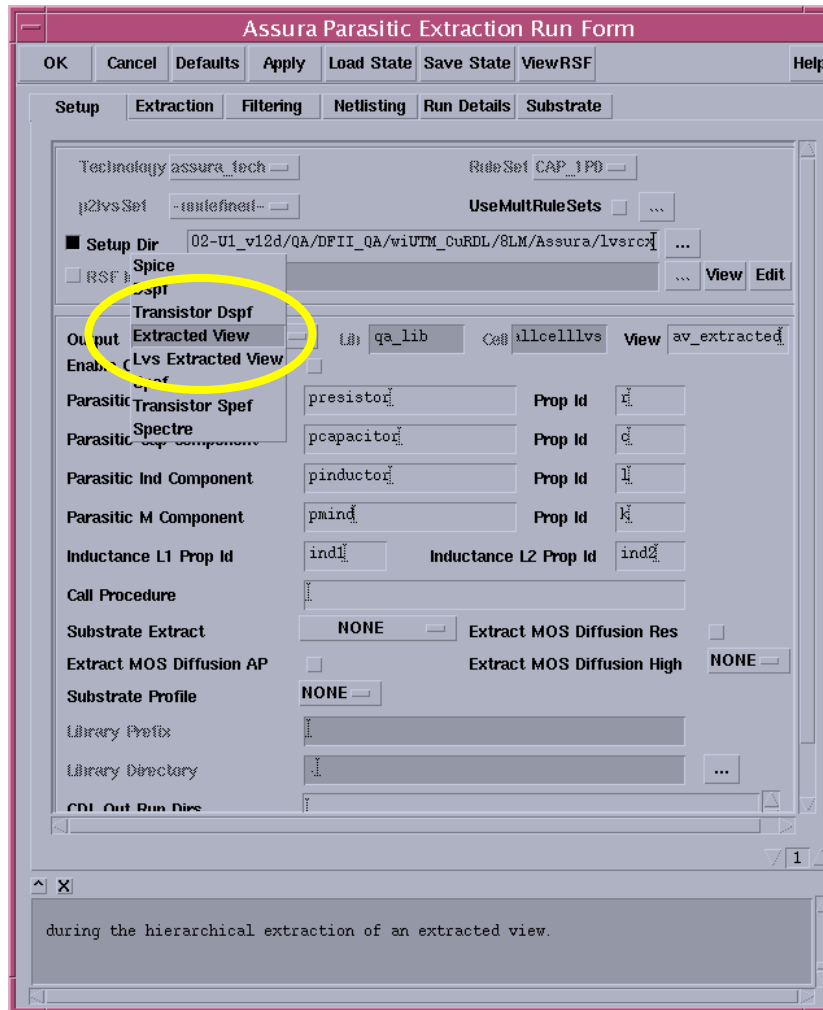


Assura LPE GUI Flow(II)



- Please turn off the “enable CellView Check” when do the LPE extraction in the “setup” folder.

Assura LPE GUI Flow(III)



- Select “Output” to “Extracted View” in “setup” folder of Assura Parasitic Extraction window to output extract result to “av_extracted” view.

Assura LPE GUI Flow(IV)

Assura Parasitic Extraction Run Form

OK Cancel Defaults Apply Load State Save State ViewRSF Help

Setup Extraction **Filtering** Netlisting Run Details Substrate

Extraction Mode **C Only** Name Space Schematic Names

Max fracture length: infinite microns Temperature 25.0 C

Cap Extraction Mode Decoupled Ref Node VSS

Mult Factor 1.0 Region Limit 200 Max num of Signals 1000

PEEC Mode ☐ Ladder Network ☐ Global Frequency 0 MHz

Select... User Region View... Edit...

Extraction Mode Full Chip All Nets RCXFS Extraction Mode NONE

Exclude Via Capacitance ☐ RCXFS High ☐

From File ☐ SelfFromSch

Frequency File View Edit

Enable HRCX ☐ Split Pins ☐ Split Pin Distance 5 Microns

Enter HRCX Cells: From File ☐

HRCX Cells: Specify a list of cells which appear in the output hierarchy, requires cell name, with optional view and lib names (cell, view, lib).

- Please specify the extraction mode in the Extraction folder.

Assura av_extract view

- RC av_extract view in the library.

